

Institute of Actuaries of India

Subject SA2 – Life Insurance

April 2016 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) The Financial Condition Report ("FCR") should include:

- The purpose of the report
- The assessment of the financial position of the insurer
- Any historical development of the insurer which may have implications for its future
- Comments on particular aspects of the historical development of the insurer, such as policy documents, marketing material, sales practices etc. which have led to Policyholders' Reasonable Expectations being formed with financial consequences for the insurer
- A review of the business of the insurer and the environment in which it is operating

The Appointed Actuary shall comment in greater detail on:

- Methods and assumptions used
- Any changes made to the methods and assumptions since the previous report
- Any inadequacies in the analysis due to factors such as lack of availability of data or any other matter
- Differing nature of assumptions and the extent to which those can be relied upon
- How the assumptions compare with the recent experience and explain any significant deviations
- Any specific financial characteristics of the existing business or the new business that may require special consideration

The FCR should indicate how the following items have been addressed:

- Any unusual features of the existing business, new business being written or proposed product development
- Any financially significant reinsurance arrangement
- Effects of any known or planned changes in insurer's method of operation
- Any use of derivatives made by the insurer, its purpose and its financial significance
- Post balance sheet events since the report period

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ii)

The liabilities of the participating fund are broadly equivalent to the aggregate of the policy-wise asset shares of participating policies plus the reserves for participating policies that became reduced paid-up upon discontinuance of premiums.

As such, the key focus of the investigation would be to review the Company's approach to compute the asset shares and how it compares to the actual cash flows in the participating fund. These are detailed below:

a) Acquisition Expenses

In the recent past, the Company has been writing about 60% of its new business through the bancassurance channel. It will be important to review whether the contribution of bancassurance to the participating new business is similar or vastly different as compared to the tied individual agency channel and the relative expense levels of the two channels.

If the Company charges a blended expense, reflecting a certain channel mix between bancassurance and tied agency to the asset shares, and if bancassurance is a more cost efficient distribution channel, then this would contribute to the increase in estate.

Therefore, examining the expense levels of the two channels vis-à-vis the expenses that the Company has charged to the asset shares over the last few years will help assess whether this has contributed to the estate.

b) Maintenance Expenses

The Company's approach to determine maintenance expenses charged to the asset shares needs to be studied. For instance, if the Company charges maintenance expenses based on what was illustrated to the policyholder and if the actual maintenance expenses are significantly lower on account of the Company achieving greater economies of scale than anticipated, then this could contribute to the increase in estate. This may have been achieved if, for instance, the bancassurance distribution channel (through which the Company writes 60% of new business) has helped the Company achieve expense efficiencies.

Overall, it may be useful to review whether there has been a recent change in the expense allocation philosophy which has resulted in lower expenses (whether acquisition or maintenance) being allocated to the participating fund. If such a reduction in the expense allocation has not been reflected in charges to the asset shares, then this could contribute to the increase in estate.

c) Commissions

Ordinarily, this is relatively straightforward as the commission scales are directly charged to the asset shares. However, there are certain aspects that merit further investigation:

- What is the Company's practice on commission savings for orphaned policies i.e. policies sourced by agents who have subsequently left the Company? If these savings are not reflected in the asset share calculations, then they could contribute to the increase in estate.
- If the Company "recovers" any service tax from the distributors and if this has not been reflected in the asset shares, this could contribute to the increase in estate.
- If the Company claws back commissions from the distributors upon policy discontinuance and if such clawed back commissions are not reflected in the asset shares of remaining policies, then this could contribute to the increase in estate.

d) Investment Returns

Review the Company's approach to determining the investment returns credited to the asset shares. In particular, are there any mismatches between the investment return credited to the asset shares vis-à-vis the investment return earned by the participating fund? For instance, if the investment return credited to the asset shares reflects a smoothed return and the actual returns on the assets in the participating fund in the recent past have been more robust, then this could contribute to the increase in estate (unless offset through changes in smoothing reserve, if any).

The determination of rate of return credited to the asset shares may entail a deduction towards investment expenses. If the actual investment expenses of the participating fund are lower, then this could contribute to the increase in estate.

Even if the investment returns earned on the participating fund and that credited to asset shares are calculated consistently, mismatches in the amounts could arise if the asset share computation formula incorporates certain approximations or simplifications. For instance, if the asset shares are only calculated based on the asset shares at the start of the year with cash flows during the year being assumed to be incurred at the middle of the year, such an approximation may have contributed to the increase in estate. As such, it may be useful to review the amount of investment income credited to the asset shares vis-à-vis the amount of investment income earned.

e) Cost of Insurance

Review the Company's approach in calculating the charges to the asset shares towards the costs of providing the insurance benefits. If the Company charges cost of insurance based on smoothed, long-term incidence rates, then a favorable recent experience could contribute to the increase in estate.

f) Discontinuance Surplus

Review the Company's treatment of discontinuance (lapse or surrender or paid-up) surpluses emerging in the participating fund. Often, companies do not allocate these back to the remaining policyholders in the participating fund and merely retain these within the participating fund thereby contributing to the increase in estate.

If the Company does not allocate the discontinuance surpluses to the asset shares of the remaining policyholders, it may be useful to review the discontinuance experience as well as the levels of discontinuance surplus over the recent past. An increase in such discontinuances may have contributed to the increase in estate.

g) Other Miscellaneous Sources of Surplus

Review the Company's approach to treating other miscellaneous source of surplus such as that arising from riders attached to participating policies, or non-participating business written within the participating fund, reinsurance etc. If the Company does not allocate such surpluses to the asset shares, then these will contribute to the increase in estate. In such situations, it may be useful to examine the levels of such surpluses over the recent past.

h) Taxes

Review the Company's approach to charging taxes to the asset shares. The Company may charge taxes to the asset shares, treating the participating fund on a standalone basis. This may be a reasonable approach provided the illustrations issued to the policyholders reflect this. However, in practice, the taxes are ascertained at a Company level and therefore there could be a

divergence e.g. if the Company is incurring losses in non-participating segments it may be paying less taxes than what is charged to the asset shares. Such a divergence could contribute to the increase in estate.

i) Cost of Capital

Review whether the Company deducts any charge for cost of capital from the asset shares. If so, this could contribute to the increase in estate gradually as the cost of capital deduction reflects opportunity cost and not an actual cash outflow.

j) Cost of Guarantee

Review whether the Company deducts any charge for cost of guarantee from the asset shares. If so, then in a situation where the historical investment returns have been robust and as a result the Company does not need to establish a significant level of guarantee reserves, this could contribute to the increase in estate.

In addition, a few other items that may be investigated are set forth below:

- Whilst we know that the current statutory liabilities are broadly equivalent to the aggregate asset shares of the in force policies and the reserves for paid-up policies, it will be useful to check whether this was broadly the case in the past. If the valuation basis was overly prudent in the past leading to the statutory liabilities being significantly in excess of the asset shares of the in force policies and the reserves for paid-up policies, then the release of such excess prudence may have contributed to the increase in estate.
- An under-distribution of surplus to maturing policyholders may have contributed to the increase in estate. For instance, if the Company paid maturity benefits equivalent to, say 95% of the asset shares, then the residual 5% could contribute to the increase in estate (unless transferred to a smoothing reserve).
- If the Company has a significant number of paid-up policies, then the release of prudential margins in the reserves for paid-up policies will gradually contribute to the increase in estate over time.
- The estate has been growing over the last few years. As such, the investment income on the estate will keep increasing it further. Even more so if the Company's investment philosophy for the estate entails greater investment in risky assets like equities and such assets have performed strongly in the recent past.

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iii)

The key principle would be to ascertain the benefits to the participating policyholders as a result of the upgradation. If it can be established that the participating policyholders will benefit meaningfully from the upgradation, then it may be reasonable to charge a fair share of such upgradation to the participating fund.

Various aspects that may be considered are set forth below:

1. Detailed break-up of the upgradation costs needs to be examined. Some of these may be infrequent but unavoidable business expenditure e.g. upgrading IT hardware. It may be

justifiable to allocate a fair share of such costs e.g. based on the proportion of in-force policy counts in participating and other funds. However, one-off expenditure such as that for replacing a legacy administration system will merit a deeper investigation.

2. The core issues / shortcomings with the legacy administration system which trigger the need to replace it with a new administration system need to be understood. Some considerations could be:
 - Are there generic issues / shortcomings which affect all lines of business? Examples include underwriting capabilities, new business processing capabilities, reinsurance administration etc.
 - Are there issues / shortcomings largely affecting a specific line of business? Examples include inability to manage unit-price based guarantees for unit-linked products, calculating policy level claw-back additions for unit-linked products, calculating historical policy-level asset shares for participating business etc.
 - Is the new system needed for venturing into new channels / product lines which require more agility in administration such as group, online etc.?
3. If it can be established that the new system is required mainly to address generic inefficiencies of the existing system affecting all lines of business, then allocating a part of the expenditure to participating fund may be justified. If the new system is required solely to venture into new channels or to address gaps in respect of non-participating or unit-linked products, then the case for allocating a part of the expenditure to participating fund would be rather weak.
4. Whilst the primary trigger to consider changing the administration system may not be strongly related to participating business, there may be sufficient benefits that may accrue to the participating business from the new system. For instance, the existing administration system may be incapable of computing the non-negative claw back additions payable after a unit-linked policy has remained in force for five years and, therefore, there is a gap that must be addressed. However, the new administration system may not only address this gap but also be far more efficient in producing annual bonus statements, surrender value quotes and revival quotes for participating business. In such a case, even though the primary trigger was unrelated to participating business, sufficient benefits are demonstrated for participating business to pick up a part of the costs.
5. Once there is a broad agreement that the participating policyholders will meaningfully benefit from the upgradation and should therefore pay a fair share of the costs, the obvious question is how to determine such a fair share. Often, expense allocation is more pragmatic than scientific. As such, allocating costs based on the proportion of number of in force policies or size of reserves may be sensible. If the Company has an established cost allocation philosophy for similar expenditure or a precedent, it may be reasonable to use that as a starting point.
6. The Company may need to decide whether the expense allocated to the participating fund should be charged to the asset shares or the estate. If the Company desires to charge the costs to the asset shares, it may want to consider amortising such charges over time rather than charge it only to the asset shares of the current generation of the policyholders. In addition, charging such expense to the asset shares may have ramifications on the Policyholders Reasonable Expectations (PRE). As such, if the allocated expense is a relatively small part of the estate, it may be more appropriate to charge it to the estate than to the asset shares.

Finally, it would be appropriate to discuss the analysis and recommendations with the With Profits Committee (WPC) of the Company to ensure that the members of the WPC are comfortable with the approach adopted in this regard.

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iv)

Key safeguards implemented by the regulator to protect the interests of the participating policyholders are set forth below:

1. 90/10 Gate: The single most important safeguarding measure is the entitlement of the participating policyholders to at least 90% of the distributed surplus. Furthermore, if the participating fund is in deficit, shareholders need to inject capital to eliminate the deficit in the participating fund. Such capital injections once made are irreversible. The ability of the company to make good the deficit in the participating fund and declare bonuses to the policyholders through transfers from shareholders' fund is also permitted only for the first 12 years of operations of the company, thereby forcing the company to ensure that the participating business becomes self-sufficient during this period through. Also, shareholders do not have recourse to alternative mechanisms to injecting capital such as granting a contingent loan to the participating fund.
2. With Profits Committee (WPC): Regulations require a life insurance company to constitute a WPC comprising an Independent Director of the Board, the CEO, the Appointed Actuary and an Independent Actuary. The terms of reference for the WPC entails approving the asset share calculations, including the expenses charged and the investment income credited to the asset shares. The WPC provides an additional layer of oversight for the participating funds. While the regulatory terms of reference are somewhat narrow, often the WPC members may have an over-arching regard to the interests of participating policyholders.
3. Expense Management: Life insurance companies are required to establish appropriate expense allocation philosophy to allocate expenses to participating and other funds. Furthermore, making arbitrary changes to the allocation approaches from one year to the next may invite additional scrutiny from the regulator. There exist overall limits on the expenses that an insurance company may incur, having regard to the age of the insurer and the levels of business underwritten. Recently, the regulator has proposed applying a similar sub-limit separately for the participating fund on a standalone basis which could further safeguard interests of the participating policyholders.
4. Prudential investment norms are prescribed to ensure that participating fund assets are invested in a balanced portfolio and are not overly exposed to risky assets such as equities. Even within debt investments, certain minimum allocations towards the government bonds are prescribed which controls the credit risk exposure.
5. Gross investment yields to be used for the purpose of point of sale benefit illustrations issued to the participating policyholders are prescribed by the regulation. As such, this prevents a situation where an insurer may illustrate higher benefits backed by higher levels

of risky assets such as equities without the policyholder fully appreciating the fact that the higher expected return comes with a greater volatility in returns.

6. Reserving and solvency margin requirements provide additional safeguards. Reserves are required to reflect guaranteed benefits including declared bonuses, the non-guaranteed anticipated future bonuses and also the taxes and shareholder transfers associated with such bonuses. Up-fronting of profits is restricted due to the surrender value floor on reserves, applied at a policy level.

[8]

[50 Marks]

Solution 2:

i) The main objective of asset liability management (ALM) studies is to derive an investment strategy that takes into account the insurer's attitude to market risk, meaning the extent to which it is prepared to take the risk of not being able to pay its claims due to adverse economic and investment market conditions. This attitude will vary according to how strong the insurer is in terms of the size of its available capital, and its competitive positioning within the market.

ALM studies aim to examine alternative investment strategies so that they can be assessed as to their suitability and level of risk. These could incorporate both physical asset and derivative strategies. This helps to determine, for any given investment strategy, whether there is enough money in total to meet liabilities as well as whether cash-flows are such that money is available at the right time.

[2]

ii)

Regulatory requirements for asset liability management and stress testing for life insurers are set out in a circular by the IRDAI titled "Asset-liability management and stress testing" dated 3 January 2012.

As per this circular, the following requirements apply for asset liability management and stress testing that would apply to the non-participating product specified in the question:

1. Details of the asset liability management activities undertaken by the insurance company should be reported in the Appointed Actuary's Annual Report; and must include a description of the manner in which the risk of mismatch between the assets and liability cash-flows is monitored, together with disclosure of the projected cash-flows for assets and liabilities for each line of business separately (and within a line of business, special products that are interest sensitive should be shown separately);

2. All insurers must put in place effective procedures for monitoring and managing asset liability positions and establish an ALM policy to be approved by the Board of the insurer, after having considered:
 - The relationship between assets and liabilities of the company;
 - Insurer's overall risk tolerance;
 - Risk and return requirements of the insurer;
 - Solvency position of the insurer; and
 - Liquidity requirements.

The Board is required to review the policy at least at each year end, or more frequently if necessary.

3. Disclosures with regards to asset and liability amounts, categorised by duration is required to be submitted on a quarterly and annual basis in prescribed formats.
4. The following minimum stress tests are required to be conducted and impact on assets and liabilities assessed, for stresses applied separately over one year and three year periods (i.e. two scenarios):
 - Fall of 30% p.a. in equity values;
 - 100 basis points p.a. fall in yields available on fixed interest securities;
 - Adverse deviation of 10% p.a. in mortality and morbidity;
 - Adverse deviation of 10% p.a. in expenses;
 - Adverse deviation of 10% p.a. in lapse/withdrawal rates; and
 - 25% increase and decrease in new business volumes.
5. In addition to the above, the IRDAI has set out a list of minimum risk factors that should be considered when undertaking scenario analysis and stress testing. These include:
 - Adverse deviation in mortality and morbidity;
 - Yield curve
 - Parallel shifts to the entire curve;
 - Parallel shifts for only duration greater than five years;
 - Parallel shifts for only duration less than five years;
 - Change in credit spreads
 - Equity fall
 - Market based equity crash ;
 - Stock specific event risk
 - Increase in expenses;
 - Adverse deviation in termination rates;
 - Fall in new business
 - Shock scenarios / scenarios of deterioration of company's experience arising from changes in:
 - Bonus scales (applicable for the three year scenario only);

- Reinsurance ceded;
- Reserving basis;
- Exercise rate of policy options;
- Distribution to shareholders; and
- Taxation
- Any other factors that may have significant relevance to the insurer's business may also be considered.

[6]

iii)

In order to undertake a cash-flow matching exercise, expected cash-flows need to be projected for both assets and liabilities.

Regarding asset cash-flows, we have been informed that the Company invests mostly in fixed interest government securities and corporate bonds. Given this, the considerations to be adopted for determining a set of best estimate assumptions required for projecting the asset cash-flows are as follows:

- For existing government securities used to back the liabilities as on the date of the investigation, the future coupon and maturity cash-flows can be projected for each security based on the asset data available, without requiring any further assumptions.
- In the case of corporate bonds, cash-flows may need to be adjusted for a best estimate default assumption. This may be based on an assessment of the probability of default on a security-by-security basis, or by assessing the default probability based on the credit ratings of the securities held, where available.
- For different investment strategies being tested, further assumptions may be required in respect of:
 - Expected asset characteristics and return thereon, for reinvestment or "new investments" in the future;
 - Timing and amount of any disinvestments, in case certain securities are not intended to be held-to-maturity under a given investment strategy.

The liability cash-flows comprise expected outgoes for the product in the form of policyholder benefits on death, survival and withdrawal, net of premium; and other outgoes such as expenses and commissions. The general principles that should be adopted in setting out the best estimate assumptions for projecting the liability cash-flows are:

- The best estimate assumptions should be considered separately for each homogenous groups of policies where sufficient evidence exists for differences in expected experience based on the homogenous categories;
- The assumptions should be realistic and be capable of justification;

- The assumptions should be guided by recent past experience, but may be modified by any knowledge of or expectations regarding likely future outcomes;
- Assumptions should be internally consistent;
- The best estimate assumptions used for cash-flow matching should be checked for consistency with similar assumptions used for other reporting (such as embedded valuation, pricing, reserving assumption before incorporating any margins for adverse deviations or MADs etc.)

Specifically, in the case of the non-participating anticipated endowment product, considerations with respect to the following assumptions would be necessary to set out the best estimate liability cash-flows:

Mortality

- As a starting point, using the standard industry mortality table – IALM 2006-08 may be considered and this may be adjusted in light of the available experience and expected future mortality;
- The product has been selling for a number of years, so it is likely that there would be some information available for the recent mortality experience;
- Further consideration should be given to whether there is material and credible information with respect to variations in expected mortality rates based on gender / smoker status / policy duration (particularly, if mortality is significantly different in early policy years) / distribution channel / case size etc.
- The chosen best estimate assumptions should be relevant to the underlying business mix – for example, considering the target customer base etc.
- Company's underwriting practices should also be considered – for example, if certain class of sub-standard lives are accepted on the basis of higher premium, then the best estimate assumptions should also reflect a loading for higher-than-average expected mortality for such lives.
- Where experience information is not credible, a reasonable assessment of expected future mortality rate may be made by looking at the best estimate assumption used at the time of pricing, the Company's mortality experience for other similar products, the general industry experience, mortality rates implied by the reinsurance premiums and/or use of expert judgement.
- Consideration should also be given to the expected changes in mortality over time, for example whether future mortality improvements should be allowed for, if material and relevant.

Morbidity assumption is not relevant, as based on the information provided there are no benefits that are dependant on morbidity rates.

Persistency

- In this product, the premiums are payable for a limited number of years (10) and the benefits are payable over the subsequent 15 years. The persistency assumption should reflect this nature of the product and may be set differently for policy years during the premium payment term and thereafter.
- As in the case of mortality, the available experience information should be considered in setting the assumption, together with any adjustments necessary based on credibility of experience rates, overall Company and industry level persistency trends and related considerations. The expected impact of any specific initiatives undertaken by the Company aimed at improving persistency may also be considered.
- It is likely that persistency rates vary by duration. In determining such rates, further consideration should also be given to the available surrender value based on duration, changes therein from year to year, and how the surrender value scale may impact the policyholder behaviour. For example, specific durations where there are likely to be one-off “spikes” in withdrawal rates should be given special attention.
- Aside from persistency rates varying by duration, may consider whether rates varying by factors such as premium mode, case size, distribution channel etc. are also relevant.
- In addition to policy discontinuance based on surrender rates, consideration should be given to whether premium discontinuance (for example, by making the contract reduced paid up) is also relevant.
- While determining the overall persistency assumption, any revival or reinstatement options (and duration of the revival period) should also be considered. For simplicity, future discontinuance rates may be assumed net of revivals.
- The available surrender values are likely to be less or more attractive depending on the prevailing economic conditions. Therefore, in setting out the persistency assumptions, care needs to be taken to ensure that these are consistent with the underlying economic scenario projected. In case considering different scenarios for testing different investment strategies, consideration should be given whether persistency assumptions should also be changed dynamically based on the likely policyholder actions under the specific scenario.

Together with the information on the product features; policyholder details from policy data and the above assumptions in respect of mortality and persistency, it is possible to set out the cash-flows for the expected premium income and policyholder benefits payable under the product. In addition to these, the cash-flow projection should also incorporate expenses and commissions as part of the liability cash-flows. Following assumptions would be required in this case:

Expense and expense inflation

- The level of expenses considered for the purpose of cash-flow matching should be at the level that is expected to be met by the assets backing the liabilities and future premiums. Any excess, for example, expense over-runs that are not intended to be borne by the

underlying assets and future incomes may need to be excluded for cash-flow matching purposes.

- In addition to having regard to the recent expense investigations and the Company's expense analysis, the assumed unit cost loadings (per policy, % of premium or % of sum assured related etc.) should also be checked for consistency with corresponding loadings implied by the projected expenses and business volumes as per the Company's business plan.
- Similarly, expense inflation should be consistent with the expected increases in expenses – these should consider both wage inflation (for salary costs), and the general inflation in the economy for the Company's other expenses.

Commissions

- Assumption for commission outgo should be consistent with the actual commissions payable to intermediaries. As a starting point, the commission scale mentioned in the 'File and Use' document at the time of product launch may be considered.
- Adjustments may be required for any variance in actual commissions from that mentioned in the 'File and Use' documents. For instance, in case of certain distributors, if no further commissions are paid (due to termination of their distribution contracts or attrition), these should be allowed for while setting the overall assumption for renewal commission.
- Additionally, any channel specific features should be allowed for – for instance, in case of policies sold through 'direct' channel, no commissions might be applicable. In this case, the cash-flows should ideally allow for this through explicit modelling. Alternatively, if allowance through modelling is not possible, then appropriate adjustments might be necessary in the assumed commission rates, taking into account the proportion of business sold through such channels.
- Wherever applicable, allowance should be made for commission clawbacks based on the internal rules set by the Company.

Other items

- Further consideration might be given to whether any other cash-flows are relevant for the product. For instance, if the cash-flow projection is undertaken to include allowance for reinsurance cash-flows, appropriate assumptions might be required to project reinsurance cash-flows, which should ideally reflect the terms of the reinsurance treaties. In particular, reinsurance premiums may be projected based on the premium rates charged by the reinsurer, allowing for any applicable discounts; and reinsurance outgoes should be based on the assumed mortality rates and ceded benefits.
- It might be unusual to allow for taxation in an ALM cash-flow matching exercise, particularly when doing so at an individual product level, in which case no assumption for future tax may be necessary.

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iv)

The 'Option B' for this product represents an embedded option with the investment. For estimating the cost of such a policyholder option and investment guarantee, a stochastic model may be recommended.

The following components would be required for building such a stochastic valuation model:

Asset model and an Economic Scenario Generator (ESG) providing stochastic investment returns scenarios

The policyholder benefits under Option B are linked to the returns earned on the underlying invested funds. If future investment returns (and corresponding crediting rates) exceed a certain level such that the maturity benefits are more than the sum of the fixed annuity that would have been payable under Option A, then there is no cost to the Company. However, in case the accumulated funds are insufficient to provide for the minimum guaranteed value as payable under Option A, the Company has to fund the difference resulting in a cost to the Company. The lower the future investment returns, the higher would be such cost to the Company. Hence, a range of future investment returns scenarios need to be tested. This may be projected using a stochastic model that provides such large number of investment returns scenarios.

At the outset, we would need to make a decision as to whether the stochastic projection should use "real world" or "risk neutral" measure for investment returns. Given the purpose of the model is to estimate a cost of guarantee, risk neutral scenarios may be preferable.

The stochastic projections would model a central projection of the investment returns representing the mean outcome, together with volatility of returns around the mean outcome to allow for the different possible scenarios.

The number of scenarios to be projected should be large enough to reduce or eliminate sampling error while being practically manageable.

The investment returns would be modelled separately for government bonds, corporate bonds and money market instruments (may be assumed to be cash). The model should allow for consolidating total returns from individual asset classes to a weighted portfolio return for each time-step. The weights to be applied would be determined by the current and an assumed future asset mix, relevant within each scenario.

In the case of a "risk neutral" projection, the 'certainty equivalent' projection would assume a risk free return for all asset classes, and differences in returns for different asset classes may be modelled through volatility of returns for the corresponding asset class within the stochastic scenarios. The certainty equivalent risk free return may be taken as the government bond yield curve as at the calculation date and volatility assumption may be determined based on the market implied swaption volatilities. For corporate bonds, the credit default risk may be

modelled within the stochastic total returns index via the implied volatilities for the corporate bonds. The risk free forward rate under each scenario may be used for discounting cash-flows in the model.

In case a “real world” stochastic projection of returns is chosen, then the mean returns would represent the best estimate of expected real world total returns for each asset class. The volatility of returns would also reflect the expected real world distribution of returns. Returns on corporate bonds may be modelled via an explicit spread over the risk free rates. In the case of real world projections, a set of state price deflators would be required for discounting cash-flows in the model. As noted previously, for the purpose of calculation of cost of option and guarantees, risk neutral scenarios may be considered more appropriate.

The output of the asset model would be the weighted average portfolio returns for each time-step under each scenario based on the underlying asset mix and stochastic asset returns. These would interact with the liability cash-flows and other model routines to produce the required cost of policyholder options and investment guarantee.

Liability cash-flow projections

A cash-flow liability model would be required to project the premiums and policy benefits. The liability cash-flows may be projected policy by policy or in case there are other practical constraints such as the run times or computing power / memory, then grouped model point data may also be used.

The liability model would comprise:

- **Modelling of independent decrements** – These would be based on the best estimate assumptions for mortality and persistency. Generally, it might be sufficient to model these on a deterministic basis, as the primary focus of the current exercise is to calculate the cost of policyholder option and investment guarantees.
- **Dynamic policyholder behaviour** - Some assumptions may need to be dynamically linked to the investment scenario (as these represent the underlying economic conditions for the particular simulation) – for example, we may wish to consider dynamic policyholder behaviour with respect to:
 - Persistency, particularly since guaranteed surrender values may be available.
 - Option take up rate, as the description for the policyholder option suggests that this is available at the end of the premium payment term.

Consequently, these assumptions should be capable of being linked with the investment scenario through an appropriate algorithm in the model.

- **Modelling of dependent decrements** – Dependency of different policy “states” should be allowed for in the model, particularly with respect to interactions between mortality rates, lapse / surrenders and premium persistency.

- **Accumulation of asset values and future premiums–** For each policy, the market value of base backing assets excluding the assets allocated to the liability in respect of the embedded options and guarantee, are accumulated at the simulated investment returns together with future policy cash-flows. The future policy cash-flows would allow for an appropriate probability measure and consist of the following aspects:
 - Fixed, regular premiums, projected for up to ten years after date of inception. Applicable premium modes should be allowed for.
 - While projecting the available funds at maturity, cash-flows in respect of benefits payable upon death, surrender as well as other policy outgoes such as expenses and commissions should be deducted from the accumulation of starting assets and premiums.

Accumulation of the starting value of assets plus future premiums less outgoes provides the value of available funds at maturity.
- **Estimation of (variable) maturity benefit under Option B:** In order to compute the maturity benefit under Option B, annual crediting rates may be modelled using an appropriate investment management spread over the earned investment return on the fund (available from the output of the asset model). Such a spread or some other defined relationship between earned investment return and crediting rates may also be pre-specified in the product literature.
- **Allowance for management actions:** Possible management actions should be allowed for in the scenario projections. Some examples of management actions that could be considered are rebalancing of the asset portfolio by changing asset mix depending on the investment conditions (as implied by the simulated returns) or changing the amount or pace at which crediting rates are declared or changing the investment management spreads (unless they are contractually defined). Ideally, any such management actions modelled should be well documented and based on approved Company policies.

Estimation of the cost of the option and investment guarantee

Based on the above accumulations, there would be three values that could be compared at the end of the maturity period under Option B:

- i. The accumulated asset value available to pay benefits upon maturity;
- ii. The accrued maturity benefit based on crediting rates declared during the 15 year pay-out period; and
- iii. The minimum guaranteed maturity value defined as the sum of the fixed annuity that would have been paid under Option A.

The maturity benefit under Option B is defined as the higher of the accrued benefit and the guarantee benefit (i.e. Higher of (ii) or (iii) above).

If the accumulated asset value under (i) above is greater than the maturity benefit so derived, then there is no cost to the Company. However, in case of a shortfall, the difference between the two values represents the cost under the particular scenario. The shortfall at maturity date may be discounted back to the valuation date.

The above calculation would be repeated across all stochastic simulations resulting in a distribution of results representing the present value of the shortfall. The average of these would be taken as the cost of the policyholder option and investment guarantee.

[16]

v)

The following checks should be applied on the model built to estimate the cost of the policyholder option and investment guarantee:

1. Check that the model code is robust and test for computational integrity of the model. This may be done by a close inspection of the model code or by developing independent calculations to reconcile against the model results on a sample basis;
2. Validate all model inputs, including data, assumptions and other run parameters;
3. Validate all assumptions for consistency against data from other business units – for example, asset mixes, expected future mean returns and liability assumptions against other actuarial applications;
4. For stochastic investment returns, check that the martingale and convergence tests are satisfied (and in case of risk neutral scenarios, no arbitrage assumption holds true for the scenario returns);
5. Check simulated returns for consistency – for example, check for correlation among different asset returns across scenarios, internal consistency of assumptions under different simulations etc.
6. If sample model points from the policy data are used, check for any sampling error and ensure that the model points are sufficiently representative. This may be done, for example by checking that a different choice of model points within similar homogenous policy groups do not result in significantly different results.

In case model points are created based on portfolio averages (rather than being a selection of some sample policies within the policy data) then these may be tested by comparing results from the aggregate policy-by-policy portfolio run against the model points results (say, for a smaller number of scenarios or a few deterministic scenarios only);

7. The scaling up of the model points should also be checked – for example, ensure that the total number of policies or starting asset values are consistent between scaled up model point results and aggregate policy data;

8. Check for reasonableness and robustness of results – for example, by running a small set of deterministic scenarios. One can then check that the cost is zero for high investment returns, and that the cost increases as the future investment return assumption reduces;
9. Check that the average across various scenarios of the present value of the difference between (i) the accumulated asset value available to pay benefits upon maturity and (ii) the accrued maturity benefit based on crediting rates declared during the 15 year pay-out period is comparable to the investment management spread;
10. Another sense-check may also be carried out by running the model to solve for a “break-even” investment return, which could be tested against pricing assumptions and overall reasonableness;
11. Test boundary conditions in the model – particularly in respect to any dynamic policyholder behaviour; logic used for crediting rates or management actions modelled. These could be done by testing extreme scenarios to ensure minimum and maximum ranges for these modelled features are not breached;
12. Further reasonableness checks may be considered such as:
 - a. Expected guarantee cost as a percentage of premium or as a percentage of the present value of premiums
 - b. Movement in guarantee cost from prior results or pricing (if available);
 - c. Movements in results from a prior model, if such a model exists;
13. Sensitivity analysis should be performed, particularly for key model inputs (e.g. volatility of asset returns used in the stochastic projections / asset mix / policyholder behaviour etc.). Results from these should be checked for reasonableness, both in terms of directional as well as amount impact. Any results that are not intuitive should be investigated further as these may reveal coding or parameter input errors.

[10]

[50 Marks]
